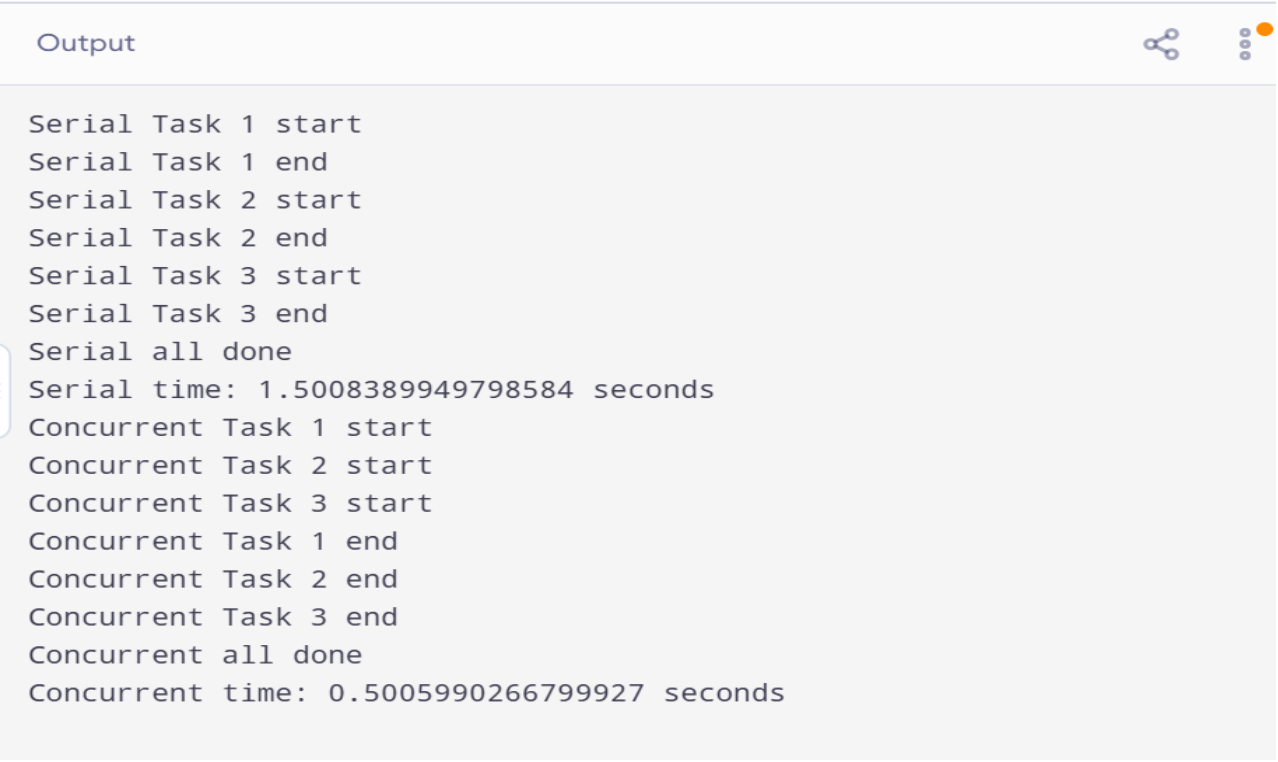
Task 1: Guided Activity, Serial versus concurrent

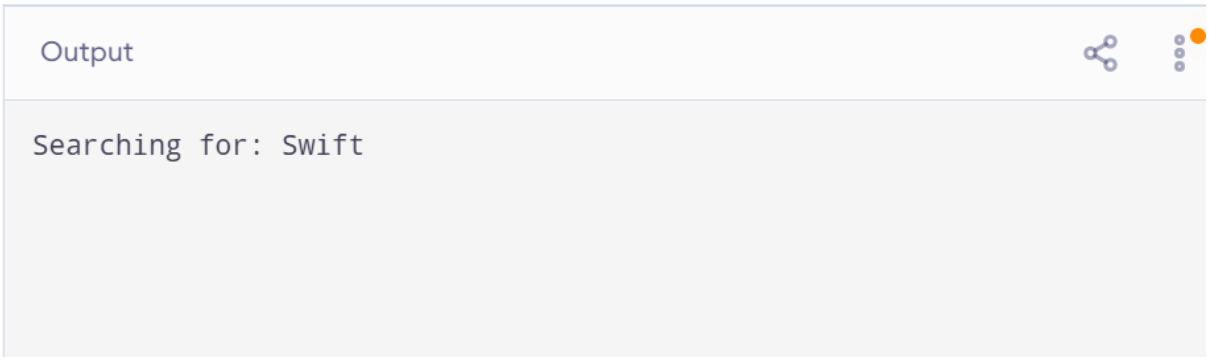
```
import Foundation
let serialQueue = DispatchQueue(label: "serialQueue")
let concurrentQueue = DispatchQueue(label: "concurrentQueue", attributes: .concurrent)
let group = DispatchGroup()
let startTime = Date()
let serialGroup = DispatchGroup()
for i in 1...3 {
   serialGroup.enter()
   serialQueue.async {
      print("Serial Task \(i) start")
      Thread.sleep(forTimeInterval: 0.5)
      print("Serial Task \(i) end")
      serialGroup.leave()
   }
}
serialGroup.wait()
print("Serial all done")
print("Serial time: \(Date().timeIntervalSince(startTime)) seconds")
let concurrentStart = Date()
for i in 1...3 {
   group.enter()
   concurrentQueue.async {
      print("Concurrent Task \(i) start")
      Thread.sleep(forTimeInterval: 0.5)
      print("Concurrent Task \(i) end")
      group.leave()
   }}
group.wait()
print("Concurrent all done")
print("Concurrent time: \(Date().timeIntervalSince(concurrentStart)) seconds")
```



Task 2: Practice Task, Debounce a search box

```
import Foundation
class SearchDebouncer {
    private var workItem: DispatchWorkItem?
    private let queue = DispatchQueue(label: "searchQueue")
    func search(_ text: String) {
        workItem?.cancel()
        let newWorkItem = DispatchWorkItem {
            print("Searching for: \(text)")
        }
        workItem = newWorkItem
        queue.asyncAfter(deadline: .now() + 0.4, execute: newWorkItem)
    }
}
let debouncer = SearchDebouncer()
```

```
debouncer.search("S")
Thread.sleep(forTimeInterval: 0.1)
debouncer.search("Sw")
Thread.sleep(forTimeInterval: 0.1)
debouncer.search("Swi")
Thread.sleep(forTimeInterval: 0.1)
debouncer.search("Swif")
Thread.sleep(forTimeInterval: 0.1)
debouncer.search("Swift")
Thread.sleep(forTimeInterval: 1.0)
```



Task 3: Debug a Race Condition in Swift

```
import Foundation
var count = 0
let lock = NSLock()
DispatchQueue.concurrentPerform(iterations: 1000) { _ in
    lock.lock()
    count += 1
    lock.unlock()
}
print(count)
```

Output





```
1000
```